

LAB - Deserialization

LAB - Deserialization - Author not stated, HackMD.

- Published: date not stated
- Original: <https://hackmd.io/@meowhecker/ryd4Jz17A>
- Preserved from: <https://hackmd.io/@meowhecker/ryd4Jz17A> (preserved-copy) on 2026-08-04
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HackMD - Collaborative Markdown Knowledge Base

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[]()LAB-1: Sub-domain / Path Traversal -> Download web.config / Insesurity Deserialization - viewstate **ASP.NET / Powershell credential Cracking / runas.exe / seDebugPrivilege abusing**

<https://app.hackthebox.com/machines/585>

[]()Recon

[]()Enumerate Attack surfaces

[]()SubDomain

[]()Identify & Exploit Vulnerabilities

[]()ASP.NET __VIEWSTATE

__VIEWSTATE is used to management page and control data cross the web page, During the rendering of a page HTML.

[Exploit - Reference](#)

[]()Case

EnableViewStateMac = True/False (?) ViewStateEncryptionMode = True

We need machine key to inject object chain to invoke systems call

Machine key typically was stored in web.conf

we need to find the endpoint to do path traversal to steal web.conf

[()]Find out web.conf

Fuzzing Sensitive File

Testing /web.conf

Very lucky, developer didn't ViewStateUserKey setting !

Machine Keys

[()]Exploit (ysoserial.exe)

Basic .Net Deserialization Construct Object chain via ysoserial.exe

Modify

- path

Work

[()]Base 64 powershell to RCE

Reverse Power shell

[()]Privilege to Normal User

Cracking Power shell automatically connection Credential !

Work

[()]Runas.exe (Local privilege to normal user)

Download Runas (Anti-virus Bypass)

[()]Privilege to root user

[()]Enable seDebugPrivilege (psgetsys.ps1,EnableAllTokenPrivs.ps1)

Run Script in Powershell

Powershell

Upload meterpreter shell to Target systems for migrate cmd to admin process !

[]()LAB-2: / Reset Password logic flaw/web.config leak via File upload, shtml/SSRF - Access Sensitive Decrypt endpoint / deserialization - viewstate **ASP.NET**

<https://app.hackthebox.com/machines/Perspective>

[]()Recon

Domain Name -> <http://perspective.htb/>

[]()Identify / Exploit vulnerabilities

Register Account

Forget The Password

Support

-> Admin Username

[]()Password Reset logic flaw to Admin

Website will check first step password reset

When we want to reset the password /Account/Forgot will stop us action

Developer didn't check emailhider parameter -> It allow attacker arbitrary specify user to reset password

Reset Password

Query All user product

Viewstate -> Probably contain Deserialize vulnerability !

[]()File upload to steal sensitive file (shtml,shtm)

We need to steal web.config

[]()Upload bypass

- Content-Type:
- Extension Name /FileName Control ?
- Magic String
- HTTP Verb

Content-Type filter Detected !

Extension Name filter Detected !

Check Extension is blacklist .

Attempt upload file to steal sensitive file.

[]()Fuzzing valid Extension

[]()shtm,shtml include web.config

shtm.shtml ->server side include (they are extension of html that allow dynamic include the file and embedding it on page)

We can attempt construct the malicious shtm,shtml to include sensitive file.

Attempting

steal.config

Machine Key we get

Analysis web.config

[]()ViewStateUserKey Exists !

Website have ViewStateUserKey setting to protected the Deserialization attack

we have to find a way to decrypt "ENC1:3UVxtz9jwPJWRvjd11PfqXZTgg=="

we also discover the sensitive port -> 8000

[]()SSRF to access Sensitive API endpoint

SSRF to enumerate internal Port

Testing Filter

Description allow to using '<'

[]()Image tags Attempt

[]()meta tags Attempt

Filter Bypass

[]()SSRF - Exploit (API endpoint Analysis)

It will return the response via xml (we have to use source code to look at)

Insert payload to Access sensitive Endpoint

Reading Document

/swagger/v1/swagger.json

<https://swagger.io/specification/>

Using Endpoint via SSRF

To do this, we can let server to fetch our website and perform CSRF Attack to Using sensitive endpoint

Goal -> Decrypt -> ViewStateUserKey

```
enc1:3UVxtz9jwPJWRvjdl1PfqXZTgg==
```

[]()Via javascript to passing parameter (Decrypt viewstateuserkey)

/encrypt?plaintext=meowhecker

meowhecker

It look like ViewStateUserKey `<add key="ViewStateUserKey" value="ENC1:3UVxtz9jwPJWRvjdl1PfqXZTgg==" />`

/decrypt?cipherTextRaw

Alternate Payload

Key

[]()RCE Via Deserialization

Listener - Interface `sudo tcpdump -ni tun0 icmp`

RCE (Windows Powershell)

[]()Reverse PowerShell (Base64)

<https://github.com/itm4n/PrintSpoofer>

<http://10.10.14.5:80/PrintSpoofer64.exe>

Download Bypass